Graph Neural Networks and Their Applications in Single-Cell RNA Sequencing

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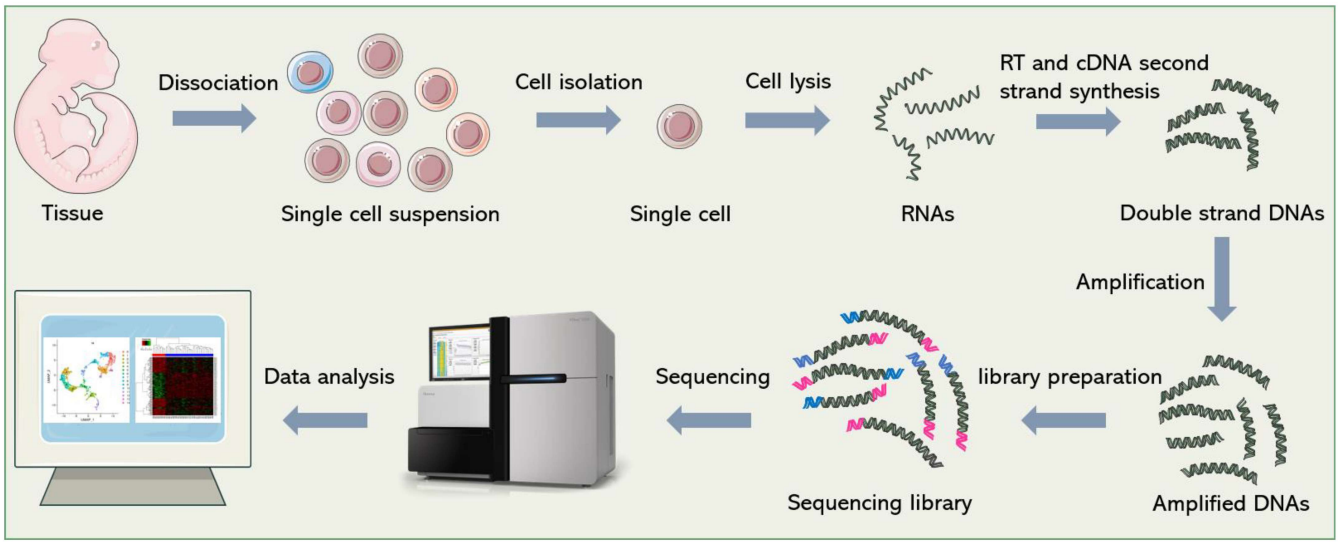

Figure 1. scRNA-seq pipeline diagram
Image Source: Online Encyclopedia

Abstract

Single-cell RNA sequencing (*scRNA*) is a modern sequencing technique that can resolve gene expression to the level of individual cells. This allows for a deeper and more thorough understanding of the underlying biology and has many applications in research, diagnostics, and fundamental research. However, studying scRNA-seq data has many associated challenges, such as high dimentionality, high sparsity and high noise. Hence, to find meaningful information from this technique, machine-learning methods have been developed that are able to perform operations such as clustering, to identify cell populations of interest, for further study.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Clustering**; *Neural networks*; *Bioinformatics*.

Keywords: single-cell RNA sequencing, clustering, graph neural networks, single-cell analysis, bioinformatics, unsupervised learning

1 Problem Statement and Importance - Introduction

Single-cell RNA sequencing (scRNA-seq) is a powerful technique that enables the measurement of gene expression at the resolution of individual cells [2]). Unlike bulk RNA sequencing, which averages expression across large populations of cells, scRNA-seq reveals cellular heterogeneity within complex biological systems. This allows for a variety of downstream applications in fundamental research and medicine.

However, scRNA-seq datasets are typically high-dimensional, sparse, and noisy, making computational analysis challenging. A core requirement to make sense of the data, is the need to properly bucket cells, on the basis of expression profiles, cell properties etc.

Hence, **clustering** is a fundamental step in almost all scRNA-seq analyses, and developing a good clustering technique will have strong downstream impact. **Clustering** facilitates downstream analyses such as marker gene identification, trajectory inference, and the study of cell–cell interactions. Consequently, improving clustering algorithms for scRNA-seq data remains an important research problem in computational biology.

Most modern scRNA-seq analysis pipelines construct a cell similarity graph and apply graph-based clustering algorithms to detect communities of related cells. Recently, machine learning approaches such as graph neural networks have been proposed to better capture relationships between cells and improve clustering accuracy. In this project, we study such graph-based approaches for clustering single-cell RNA-seq data and evaluate their effectiveness on publicly available datasets.

In this project, we intend to use scGNN ([5]) a graph neural network based method, to classify different cell types based on their RNA expression profiles, from publicly available scRNA-seq datasets.

2 Topics Covered and Subtopics

This project focuses on computational methods for clustering single-cell RNA sequencing (scRNA-seq) data, with an emphasis on graph-based approaches and graph neural networks. The following topics will be covered in the lecture and explored through experiments on publicly available datasets.

2.1 Single-Cell RNA Sequencing Data

We begin with an overview of single-cell RNA sequencing and the characteristics of the data produced by this technology. Topics include the structure of scRNA-seq expression matrices, technical challenges such as sparsity and dropout events, and the importance of identifying cell populations within heterogeneous tissues.

2.2 Clustering in Single-Cell Analysis

Next, we discuss the role of clustering in single-cell analysis, followed by a brief discussion of different non-GNN based approaches described in literature such as statistical methods and k-means.

2.3 Graph-Based Clustering Methods

Finally, we examine recent machine learning approaches that use graph neural networks (GNNs) to improve clustering performance, by describing the architecture of scGNN [5], the salient features, and implementation

3 Software and Datasets Used

3.1 scGNN (single-cell Graph Neural Network

The primary tool to be evaluated in this study is scGNN [5], a graph autoencoder-based deep learning framework for unsupervised scRNA-seq clustering. The scGNN implementation is publicly available here.

3.2 Datasets

All datasets will be accessed through Bioconductor via the scRNAseq R package (ExperimentHub) or their respective public repositories as SingleCellExperiment objects.

3.2.1 Baron Human [1].

8,629 human pancreatic cells, 17,499 genes, 14 cell types, sequenced using the inDrop droplet-based platform.

3.2.2 Baron Mouse [1].

1,886 mouse pancreatic cells, 14,861 genes, 13 cell types, sequenced using the inDrop droplet-based platform.

3.2.3 PbmcBench pbmc1.10Xv2 [3].

6,444 PBMC cells, 33,694 genes, 9 cell populations, sequenced using 10x Chromium v2.

3.2.4 PbmcBench pbmc2.10Xv2 [3].

3,362 PBMC cells, 33,694 genes, 9 cell populations, sequenced using 10x Chromium v2.

3.2.5 CellBench CEL-Seq2 [4].

570 human lung adenocarcinoma cells from three cell lines (HCC827, H1975, H2228), 12,627 genes, 3 cell types, sequenced using the CEL-Seq2 plate-based protocol.

3.2.6 Zheng Sorted [6].

20,000 PBMC cells across 10 FACS-sorted immune cell populations, 21,952 genes, sequenced using 10x Chromium.

4 Planned Experiments

This study will benchmark the scGNN clustering architecture across three pairwise dataset comparisons, each designed to probe a distinct aspect of scGNN's known strengths and limitations. Of particular relevance is the iterative autoencoder-based hybrid loop at the core of scGNN, which becomes increasingly resource-intensive as dataset size grows, and the reliance on graph-based topological learning, which requires a sufficient number of cells to construct meaningful cell-cell relationships.

4.1 Comparison 1:Cross-species performance (Baron Human vs. Baron Mouse)

scGNN will be applied independently to both pancreatic datasets. As both originate from pancreatic islet tissue and were generated using the same inDrop platform, this comparison will isolate the effect of species-level biological and genomic divergence on clustering outcome.

4.2 Comparison 2:Effect of cell number on clustering efficiency (pbmc1.10Xv2 vs. pbmc2.10Xv2)

As both PbmcBench subsets were generated using 10x Chromium v2 chemistry and annotated with the same nine cell populations but differ in cell count (6,444 vs. 3,362 cells), this comparison will directly assess how cell number influences the quality of the learned graph embeddings and the associated computational overhead of the iterative autoencoder loop

4.3 Comparison 3: Efficiency vs. accuracy trade-off (CellBench CEL-Seq2 vs. Zheng Sorted)

This comparison will characterise the trade-off between computational efficiency and clustering accuracy across contrasting dataset sizes. The CellBench CEL-Seq2 dataset at 570 cells is expected to run efficiently but perform poorly, falling below the 1,000-cell threshold identified by the scGNN authors as necessary for the iterative autoencoder loop to learn reliable high-order cell-cell representations. The Zheng Sorted dataset at approximately 20,000 cells is expected to yield stronger clustering outcomes at the cost of significantly greater runtime and memory usage.

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